

Week 4: Alerting, Testing, and Final Reporting

The final week is dedicated to safety and visualization. The intern must implement a rollback mechanism allowing SOC analysts to reverse an automated firewall rule if a false positive occurs. They will finalize the Kibana dashboard to visualize the blocked threats and ensure the GitHub repository contains all scripts and architectural documentation.

Project 2: Healthcare - IoT Deception Honeypot Network

Executive Problem Statement

Healthcare environments are increasingly reliant on smart, connected cyber-physical systems, ranging from patient vitals monitors to smart HVAC infrastructure. Unfortunately, many of these Internet of Things (IoT) devices lack robust built-in security, making them prime targets for botnets and ransomware gangs seeking to pivot into the broader hospital network.

The objective of this project is to develop an IoT Deception Honeypot Network. The intern will architect a virtual environment consisting of simulated, low-interaction vulnerable IoT devices to proactively deceive attackers, trap them, and analyze their behavioral patterns and exploit techniques before they can reach actual medical equipment.

Business Objectives and Key Performance Indicators

The strategic goal is to shift from reactive defense to proactive threat intelligence gathering. By utilizing deception, the organization can identify reconnaissance attempts and zero-day exploits targeting specific medical IoT protocols.

Success is measured by the honeypot's ability to convincingly simulate a device, successfully log an attacker's interaction (IPs, payloads, commands attempted), and visualize these attack vectors on a centralized threat analysis dashboard without allowing the attacker to escape the sandboxed environment.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Threat Researcher (Intern)	Safe environment to capture and analyze live malware.	Deploys the honeypot nodes, monitors attacker behavior, and extracts dropped malware payloads for reverse engineering.

Network Administrator	Early warning system for internal lateral movement.	Receives immediate alerts if an internal IP address attempts to interact with the hidden honeypot, signaling a compromised internal machine.
Compliance Auditor	Proof of advanced, proactive security measures.	Utilizes the threat intelligence gathered to justify network segmentation policies required for HIPAA compliance.

Minimum Viable Product Specifications

The foundational requirement is the Virtual Honeypot Deployment. The intern must utilize tools to simulate the behavior and open ports of typical IoT devices (e.g., default SSH/Telnet credentials, unauthenticated web panels).

The core deliverable is the Logging and Analysis Engine. When an attacker connects to the honeypot, the system must meticulously record every keystroke, uploaded file, and network request. This data must be parsed and visualized to identify the origin of the attacks and the specific vulnerabilities being targeted.

Architectural Directives and Technology Stack

Component	Technology	Architectural Rationale
Honeypot Software	Cowrie / Honeyd	Open-source, low-to-medium interaction honeypots designed specifically to log brute force attacks and shell interaction.
Infrastructure	Docker / Virtual Machines	Essential for strictly isolating the honeypot from the host system, ensuring attackers cannot pivot into the real network.

Data Visualization	Python / Splunk	Used to aggregate the raw logs into a comprehensive threat analysis dashboard.
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Four-Week Engineering Roadmap

Week 1: Environment Setup and Device Simulation

The project begins with designing the deception strategy. The intern will configure an isolated Docker environment and deploy Cowrie or Honeyd. They must configure the honeypot to mimic a specific medical IoT device by altering its simulated filesystem, banner grabbing responses, and open ports.

Week 2: Exposure and Data Capture

Week two focuses on controlled exposure. The intern will safely expose the honeypot to a controlled testing network (or the open internet, if heavily sandboxed and approved). They will verify that the system is properly capturing connection attempts, login brute-forcing, and malicious payload drops.

Week 3: Log Parsing and Threat Intelligence Extraction

The third week targets data engineering. The intern will write Python scripts to parse the complex JSON/text logs generated by the honeypot. They will extract key indicators of compromise (IoCs) such as attacker IP addresses, uploaded malware hashes, and executed terminal commands.

Week 4: Dashboarding and Geolocation Analysis

The final sprint is dedicated to intelligence visualization. The intern will enhance the data with IP geolocation. They will build a dashboard visualizing the attack origins on a world map and categorizing the most frequent exploit techniques. The final GitHub repository must contain the deployment configuration, safe architectural diagrams, and a comprehensive analytical report.

Project 3: E-commerce - Enterprise IaC Pipeline

Executive Problem Statement

In the highly competitive e-commerce sector, rapid deployment is essential. However, the shift towards cloud-native architectures means that infrastructure is now defined by code (IaC). Misconfigurations in cloud templates or unpatched dependencies can instantly expose vast databases of consumer information. Traditional CI/CD pipelines that only check for software bugs are no longer sufficient.